Second-Hand Machines, Firm Capability, and Technology Adoption

Money or Know-How? Initial draft: evidence and model

Shadi Rahimfallah

November 2026

Department of Economics, York University
Supervisors: Tasso Adamopoulos, Chaoran Chen

Roadmap of the Empirical Work

	Data	Question	Analysis
Fact 1	FAT	Who buys only used machines?	LPM, sector & region FE
Fact 2	FAT	Do they use less advanced technology?	OLS on used-only / both dummies
Fact 3	Vietnam	Are all-used firms less capable, smaller?	Within-cell gaps; item-by-item; workers
Fact 4	Vietnam	Do capable firms run younger machines?	Machine age on capability $\times$ all used
Fact 5	Vietnam	Capability or wealth?	Tercile cross-tab; LPM with size fixed
Fact 6	Vietnam	Are all-used firms short of capital?	VA per machine \$; TFPR gap by tercile
Mixed	Vietnam	Who combines new and used?	Group means
Timing	Vietnam	Do all-used firms adopt less later?	Outcome at $t+2$ on machines at t

FAT: who buys used machines and what technology they use. Vietnam: capability, wealth and capital of firms that use them.

Data

World Bank FAT survey

19,332 manufacturing and service firms, 15 countries, 2019–2023

- **Machines:** bought new, used, or both in the last 3 years
- 9.8% of machine buyers bought only used; 12.4% of single-type buyers
- **Capability:** manager education, engineers, R&D workers, KPIs, incentives
- **Outcomes:** technology score (1–10), advanced technologies by function, productivity

Vietnam SME panel

Small and medium manufacturers, 2011, 2013, 2015 rounds

- **Groups:** all used ($\geq 95\%$ of main equipment bought used), all new ($\geq 95\%$ new), mixed
- 6,913 firm-rounds of 3,291 firms: 4,711 all new, 1,481 mixed, **721 all used**
- Size: log employment; wealth: log total assets; debt; equipment value and age
- TFPR $= \ln VA - 0.527 \ln L - 0.263 \ln K$ (JMP elasticities)
- All variables demeaned within sector $\times$ province $\times$ round

Reporting error: some firms switch all used $\leftrightarrow$ all new with no investment. From the 2011 transition matrix: a truly all-used firm reports all used with prob. 0.714; an adopter with prob. 0.059. Facts use reported groups; the model corrects.

Vietnam Capability Index

Item	2011 / 2013	2015
Internet use	q3d	q3e
Email use	q3ea	q3f
Website	q3f	q3g
PCs per worker	q34a	q35a
Professionals' share of workers	q102bx	q91tb/q90at
Respondent completed upper secondary	q26b	q27b
International quality certificate	q36a	q37a
Formal accounting books	panel variable	

Each item standardized within round; index = average, rescaled to s.d. 1. Cronbach's $\alpha = 0.825$.

The index measures managerial and technical know-how.

Fact 1 (FAT): Who Buys Only Used Machines?

$$\mathbf{1}\{\text{Used only}_i\} = X_i'\beta + \gamma_{\text{sector}} + \gamma_{\text{region}} + \varepsilon_i$$

Sample: firms that bought only new or only used machines. X_i: log employment, log firm age, manager education, engineers/technicians, R&D workers, KPIs, incentives, exporter, importer, foreign-owned, equipment loan, credit constraint, government support. Engineers and R&D missing for 60% / 62%: set to zero and flagged.

Dependent variable: bought only used	Coefficient	(s.e.)
Manager has college degree	-0.068^{***}	(0.011)
Number of KPIs monitored	-0.017^{***}	(0.005)
Log employment	-0.014^{***}	(0.004)
Direct importer	-0.037^{***}	(0.010)
Credit-constrained	$+0.030^{***}$	(0.010)
Observations; mean of dep. var.	9,844; 0.125	

SE clustered by country × sector (178 clusters). $^{***}p < 0.01$. Full table later.

Finance matters, but capability and size matter as much or more.

Fact 2 (FAT): Used-Only Buyers Use Less Advanced Technology

$$Y_i = \beta_U \, \mathbf{1}\{\text{Used only}_i\} + \beta_B \, \mathbf{1}\{\text{Both}_i\} + X_i'\Gamma + \gamma_{\text{sector}} + \gamma_{\text{region}} + \varepsilon_i$$

Reference: new-only buyers. Same controls as Fact 1.

	Tech. score (1–10)	Share adv. tech.	Functions adv.	Adv. fabric.	ERP softw.	Log sales per wkr.	Log VA per wkr.	Revenue TFP
Used only	$\mathbf{-0.515}^{***}$	$\mathbf{-0.014}^{***}$	-0.018^{**}	-0.028	-0.042^{***}	-0.142^{**}	-0.145^{**}	$\mathbf{-0.139}^{***}$
	(0.091)	(0.006)	(0.008)	(0.018)	(0.014)	(0.066)	(0.073)	(0.052)
	[0.001]	[0.037]	[0.117]	[0.127]	[0.054]	[0.180]	[0.135]	[0.024]
Mean, new only	6.46	0.165	0.347	0.386	0.545	10.64	9.98	0.030
Observations	12,397	12,413	12,438	4,558	12,446	8,713	7,544	6,310

SE clustered by country × sector. Brackets: wild-cluster bootstrap p by country (Webb weights, 999 draws). $^{***}p < 0.01$, $^{**}p < 0.05$.

Half a point lower on the technology scale and 14% lower TFP, even with controls.

Facts 1–2 (FAT): Full Results, Technology and Productivity

		Reg. 1	Reg. 2: technology use					Reg. 2: productivity			
		(1) Buys only used	(2) Tech. score	(3) Share adv.	(4) Funct. adv.	(5) Adv. fabric.	(6) ERP / softw.	(7) Log sales per wkr.	(8) Log VA per wkr.	(9) Revenue TFP	(10) Sales growth
Machines	**Used only**		**−0.515*****	**−0.014*****	**−0.018****	**−0.028**	**−0.042*****	**−0.142****	**−0.145****	**−0.139*****	**−0.022**
			(0.091)	(0.006)	(0.008)	(0.018)	(0.014)	(0.066)	(0.073)	(0.052)	(0.026)
			[0.001]	[0.037]	[0.117]	[0.127]	[0.054]	[0.180]	[0.135]	[0.024]	[0.416]
Size & age	Log employment	−0.014***	0.246***	0.022***	0.041***	0.062***	0.062***	−0.051**	−0.052*	−0.087***	0.019***
	Log firm age	−0.009*	0.016	0.001	0.005*	0.007	0.008	0.158***	0.137***	0.049***	−0.121***
Capability & human skills	Manager has degree	−0.068***	0.343***	0.037***	0.057***	0.052***	0.138***	0.402***	0.369***	0.168***	−0.002
	Engineers / technicians	−0.015	0.268***	−0.004	0.008	0.022	0.095***	0.192**	0.153	0.042	−0.073**
	R&D workers	0.007	0.329***	0.022	0.102***	0.090***	0.076***	0.069	0.089	−0.000	−0.069***
	Worker incentives	−0.012	0.319***	0.009*	0.037***	0.026*	0.054***	0.162***	0.191***	−0.010	−0.028
	KPIs monitored	−0.017***	0.273***	0.032***	0.031***	0.057***	0.026***	0.046	0.038	0.062***	0.016*
Trade & ownership	Exporter	−0.016	0.118	0.034***	0.056***	0.052***	0.050***	0.367***	0.406***	0.196***	0.037
	Importer	−0.037***	0.119*	0.028***	0.067***	0.108***	0.087***	0.250***	0.250***	0.105***	0.010
	Foreign-owned	−0.004	0.165**	0.036***	0.087***	0.037	0.099***	0.494***	0.407***	0.243***	−0.014
Finance & support	Loan for machinery	−0.011	−0.173***	0.006	0.010	0.009	0.011	0.090**	0.011	0.033	0.037*
	Credit-constrained	0.030***	−0.315***	−0.034***	0.006	−0.006	−0.029**	−0.208***	−0.130***	−0.069**	−0.011
	Government support	−0.034***	−0.187**	0.014*	0.044***	0.061**	0.034**	0.058	0.000	−0.067*	0.003
	Constant	0.278***	4.785***	−0.020	0.038	−0.155***	0.103**	9.699***	9.333***	−0.018	0.197
Observations		9,844	12,397	12,413	12,438	4,558	12,446	8,713	7,544	6,310	7,894
R^2		0.109	0.292	0.385	0.493	0.329	0.402	0.879	0.886	0.084	0.113
Mean of dep. var.[a]		0.125	6.462	0.165	0.347	0.386	0.545	10.637	9.978	0.030	0.109
Sector & region FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Used only, *no controls*			−0.875***	−0.055***	−0.091***	−0.161***	−0.153***	−0.373***	−0.363***	−0.163***	−0.011

(1) Reg. 1: 1 if only used, on buyers of only new or only used machines. (2)–(10) Reg. 2: reference group new-only buyers. SE clustered by country × sector in parentheses; bootstrap p in brackets. [a](1): share only used; (2)–(10): mean of new-only buyers. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Fact 2 (FAT): Outcome Variables

Technology outcomes

- **Technology score (`b7a`, 1–10):** self-reported technology level relative to other firms
- **Share of advanced technologies:** share of most-used technologies in `b8b--b15b` classified as advanced
- **Share of functions using advanced technology:** business functions with at least one advanced technology
- **Advanced fabrication:** any advanced fabrication technology (`b12a3--b12a6`)
- **Administration/planning software:** specialized or ERP software (`b8a4`, `b8a5`, `b9a4`, `b9a5`)

Performance outcomes

- **Labor productivity:** log sales and log value added per worker
- **Revenue TFP:** residual of log value added on employment and capital
- **Sales growth:** change in log sales

Fact 3 (Vietnam): All-Used Firms Are Less Capable and Smaller

$$X_{it} = \beta\, U_{it} + \mu_{s\times p\times t} + \varepsilon_{it}, \qquad \text{gap} = \beta \,/\, \text{s.d.}(X \text{ within cells})$$

X = capability index or log employment; $U = 1$ if all machines used; mixed counted with new.

	(1) Capability index	(2) Log employment
All used U (gap, s.d.)	-0.451^{***}	-0.393^{***}
	(0.043)	(0.042)
Firm-rounds	6,913	6,913

SE from 300-draw firm-cluster bootstrap. $^{***}p < 0.01$.

All-used firms: 0.45 s.d. less capable, 0.39 s.d. smaller, within sector, province and round.

Fact 3 (Vietnam): Where Is the Capability Gap?

$$\text{Item}_{it} = \beta\, U_{it} + \gamma \ln L_{it} + \mu_s + \mu_p + \mu_t + \varepsilon_{it}$$

One regression per row; size held fixed. SE clustered by firm. 6,922 firm-rounds for most items.

Dependent variable	β on all used (s.e.)
Capability index (s.d.)	-0.213^{***} (0.031)
Internet (pp)	-8.5^{***} (1.8)
Email (pp)	-5.1^{***} (1.5)
Website (pp)	-4.1^{***} (1.0)
PCs per worker	-0.045^{***} (0.007)
Owner/manager upper secondary (pp)	-7.9^{***} (2.1)
Professionals' share (pp)	-1.1^{***} (0.3)
International quality certificate (pp)	-0.2 (0.9)
Formal accounting books (pp)	-6.6^{***} (1.7)

Certificate: held by 4% of all-used vs. 7% of others; its 3.2 pp raw gap disappears with size fixed. $^{***}p < 0.01$.

The gap is in ICT, the manager's education, professionals and formal accounts.

Fact 3 (Vietnam): No Gap in Production Workers' Schooling

Employee survey merged with firms, 2009–2015

Share of production workers with	All new	All used	Gap (s.e.)
High school or more	0.596	0.637	$+0.046$ (0.038)
Technical/vocational or more	0.245	0.269	$+0.030$ (0.037)
College or more	0.018	0.014	-0.001 (0.011)

- 1,498 firm-rounds (175 all-used); same controls as main slides
- Coverage is thin: about 20% of firms, 2.6 workers per firm
- Same result when mixed firms count as new

No schooling gap on the shop floor: the gap is in know-how.

Fact 4 (Vietnam): Capable Firms Run Younger Machines

$$\text{Age}_{it} = \beta_1 C_{it} + \beta_2 U_{it} + \beta_3 \left(C_{it} \times U_{it}\right) + \gamma \ln L_{it} + \mu_s + \mu_p + \mu_t + \varepsilon_{it}$$

Age = mean equipment age (years, band midpoints); C = capability index (s.d.); $U = 1$ if all used. SE clustered by firm.

Sample	(1) All firms	(2) **All used**	(3) Any new	(4) All firms
Capability C	-0.52^{***}	$\mathbf{-1.08}^{***}$	-0.38^{***}	-0.36^{***}
	(0.08)	(0.28)	(0.08)	(0.08)
All used U				$+2.14^{***}$
				(0.24)
$C \times U$				-0.89^{***}
				(0.23)
Observations	6,913	721	6,192	6,913

(1)–(3): separate samples, $\beta_2 = \beta_3 = 0$. (4): pooled; β_3 tests the slope difference. $^{***}p < 0.01$.

Among used-machine firms, +1 s.d. capability ⇒ machines 1.1 years younger.

Fact 5 (Vietnam): Capability, Not Wealth, Predicts Used Machines

Share of all-used firms (%), by capability and wealth terciles

	Poor	Middle	Rich
Wealth = log total assets			
Low capability	16.6	17.9	14.5
Middle capability	8.3	7.7	6.0
High capability	9.0	9.5	5.6
Wealth relative to size			
Low capability	15.4	16.2	18.8
Middle capability	7.3	8.0	7.1
High capability	7.1	8.2	6.3

6,835 firm-rounds. Capability terciles of the index; wealth terciles within sector × province × round. Wealth relative to size: residual of log assets on log employment.

At every wealth level, low-capability firms are all used about twice as often.

Fact 5 (Vietnam): Wealth Does Not Matter Once Size Is Fixed

$$\mathbf{1}\{\text{All used}_{it}\} = \beta_a \ln A_{it} + \beta_L \ln L_{it} + \beta_C C_{it} + \varepsilon_{it} \qquad \text{(within-cell variables)}$$

	(1)	(2)	(3)
Log total assets	-0.021^{***}	-0.006	
	(0.003)	(0.004)	
Log employment		yes	-0.018^{***}
			(0.005)
Capability index (s.d.)			-0.031^{***}
			(0.005)
Firm-rounds	6,835	6,835	6,913

Mean of dep. var. 0.104. SE from 300 firm-cluster bootstrap draws. $^{***}p < 0.01$. Column (2) is a calibration target for the collateral parameter θ.

Richer firms use new machines because they are larger, not because they are richer.

Fact 6 (Vietnam): All-Used Firms Are Not Short of Capital

	All used − others	(s.e.)
Log value added per machine dollar	$+0.002$	(0.050)
Log TFPR gap, by capability tercile		
Low capability	-0.131^{***}	(0.042)
Middle capability	-0.162^{***}	(0.050)
High capability	-0.138^{***}	(0.048)
Interaction, all used × capability	0.000	(0.026)

Within-cell variables. Interaction: coefficient on all used × index in a regression of log TFPR on all used, the index and the interaction (6,839 firm-rounds). SE from 300 firm-cluster bootstrap draws. $^{***}p < 0.01$. FAT check: TFP gap of used-only buyers -0.12 with a degree-holding manager, -0.14 without.

Same value added per machine dollar, and the same productivity gap at every capability level.

Mixed Firms Are Adopters, Not Poorer New-Only Firms

	Capability (s.d.)	Log employment	Log assets	Log TFPR
All new (4,711)	0.00	−0.02	−0.02	+0.01
Mixed (1,481)	+0.20	+0.21	+0.24	+0.06
All used (721)	−0.40	−0.34	−0.35	−0.19

Within sector × province × round means.

Mixed firms are more capable, larger and richer: in the model they count as adopters.

Timing (Vietnam): All-Used Firms Adopt Less Two Years Later

$$Y_{i,t+2} = \beta_U \mathbf{1}\{\text{All used}_{it}\} + \beta_B \mathbf{1}\{\text{Mixed}_{it}\} + \beta_S \mathbf{1}\{\text{Self-built}_{it}\} + Z'_{it}\Gamma + \gamma_p + \gamma_s + \gamma_t + \varepsilon_{it}$$

Reference: $\geq 95\%$ new equipment. Z_{it} at t: size, assets, firm age, debt, liquidity, investment, credit constraint, trade, ownership. FE: province, sector, survey pair. SE clustered by firm.

	Technology adoption				Product innovation			Productivity and capital				
	(1) New tech.	(2) Outside tech.	(3) Imported tech.	(4) Self-built tech.	(5) Product improv.	(6) New product	(7) Either of two	(8) Log VA per wkr.	(9) Log sales per wkr.	(10) VA / assets	(11) VA / tangibles	(12) Tangibles per wkr.
Machines at t (reference: all new)												
All used	$\mathbf{-0.030}^{***}$	$\mathbf{-0.024}^{**}$	$\mathbf{-0.012}^{**}$	-0.005	$\mathbf{-0.037}^{**}$	0.004	$\mathbf{-0.043}^{**}$	-0.019	0.004	$\mathbf{-0.059}^{***}$	$\mathbf{-0.159}^{***}$	$\mathbf{0.154}^{***}$
	(0.010)	(0.009)	(0.006)	(0.007)	(0.016)	(0.007)	(0.019)	(0.022)	(0.027)	(0.021)	(0.055)	(0.045)
Size & age controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Finance & investment controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trade & ownership controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province, sector, pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,286	5,487	5,487	5,487	7,286	5,421	5,421	7,235	7,244	7,253	7,189	7,202
Mean, all new at t	0.095	0.061	0.025	0.021	0.260	0.025	0.310	10.837	12.032	0.470	0.884	11.791
All used, no controls	-0.031^{***}	-0.033^{***}	-0.016^{***}	0.003	-0.004	0.002	0.005	0.118^{***}	0.074^{**}	-0.100^{***}	-0.208^{***}	0.526^{***}

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Timing (Vietnam): Full Results

	Technology adoption				Product innovation			Productivity, capital				
	(1) New tech.	(2) Outside tech.	(3) Imported tech.	(4) Self-built tech.	(5) Product improv.	(6) New product	(7) Either of two	(8) Log VA per wkr.	(9) Log sales per wkr.	(10) VA / assets	(11) VA / tangibles	(12) Tangibles per wkr.
Machines at t (vs. all new)												
All used	-0.030^{***}	-0.024^{**}	-0.012^{**}	-0.005	-0.037^{**}	0.004	-0.043^{**}	-0.019	0.004	-0.059^{***}	-0.159^{***}	0.154^{***}
	(0.010)	(0.009)	(0.006)	(0.007)	(0.016)	(0.007)	(0.019)	(0.022)	(0.027)	(0.021)	(0.055)	(0.045)
	[0.017]	[0.020]	[0.008]	[0.193]	[0.078]	[0.622]	[0.113]	[0.197]	[0.866]	[0.020]	[0.049]	[0.090]
Size & age												
Log sales	0.010^{**}	0.004	0.005	0.005^{*}	-0.006	-0.003	-0.006	0.252^{***}	0.391^{***}	0.093^{***}	0.224^{***}	-0.029^{*}
Log employment	0.035^{***}	0.023^{***}	0.016^{***}	0.005	0.057^{***}	0.011^{**}	0.064^{***}	-0.211^{***}	-0.384^{***}	0.111^{***}	0.172^{***}	-0.493^{***}
Log total assets	0.005	0.006^{*}	0.001	-0.001	0.020^{***}	0.006^{**}	0.023^{***}	0.062^{***}	0.055^{***}	-0.186^{***}	-0.388^{***}	0.514^{***}
Firm age (decades)	-0.008^{**}	-0.006^{*}	-0.004^{**}	-0.002	-0.015^{***}	-0.001	-0.017^{***}	-0.038^{***}	-0.036^{***}	-0.012	-0.031^{*}	0.018
Finance & investment												
Debt / total assets	0.014	0.017	0.029	-0.014	0.025	0.006	0.016	0.062	0.199^{***}	-0.130^{***}	-0.104	0.062
Liquid / total assets	-0.024	-0.028	-0.008	-0.009	-0.021	-0.006	-0.031	0.192^{***}	0.155^{***}	0.238^{***}	1.165^{***}	-0.907^{***}
Investment / assets	0.000	0.007	-0.004	-0.010	0.002	-0.007	0.010	0.057	0.075^{*}	0.013	-0.007	0.037
Credit-constrained	0.018^{**}	0.011	0.006	0.007	0.005	0.001	0.008	0.010	-0.007	0.021	0.109^{***}	-0.133^{***}
Trade & ownership												
Exporter	0.033	0.011	-0.009	0.010	0.017	0.003	-0.013	-0.029	-0.057	0.066^{*}	0.121	-0.106
Importer of inputs	0.062^{*}	0.040	0.060^{**}	0.022	0.000	0.001	-0.001	-0.056	0.079	0.125^{**}	0.453^{**}	-0.325^{***}
Household business	-0.007	0.000	-0.005	-0.001	-0.008	0.005	-0.010	-0.057^{***}	-0.061^{**}	-0.020	-0.134^{**}	0.113^{***}
Direct imports (share)	-0.030	-0.007	0.040	0.009	-0.090^{**}	-0.001	-0.102^{**}	-0.000	-0.030	-0.102^{*}	-0.340^{**}	0.278^{**}
Indirect imports (share)	0.002	0.004	0.007	-0.000	-0.000	-0.000	-0.018	0.058^{***}	0.045^{*}	-0.039^{**}	-0.072	0.110^{***}
Import share missing	0.006	0.037	0.025	0.011	0.085^{*}	*omit.*	*omit.*	0.278^{**}	0.239	0.295^{***}	0.549^{***}	-0.218
Constant	-0.166^{***}	-0.110^{*}	-0.082^{**}	-0.042	-0.004	-0.030	-0.005	7.063^{***}	6.785^{***}	1.553^{***}	2.810^{***}	6.215^{***}
Observations (pairs)	7,286	5,487	5,487	5,487	7,286	5,421	5,421	7,235	7,244	7,253	7,189	7,202
R^2	0.074	0.047	0.056	0.031	0.158	0.038	0.164	0.455	0.414	0.193	0.153	0.395
Mean, all new at t	0.095	0.061	0.025	0.021	0.260	0.025	0.310	10.837	12.032	0.470	0.884	11.791
Province, sector, pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
All used, *no controls*	-0.031^{***}	-0.033^{***}	-0.016^{***}	0.003	-0.004	0.002	0.005	0.118^{***}	0.074^{**}	-0.100^{***}	-0.208^{***}	0.526^{***}

Outcome at $t+2$ on machines and controls at t. Mixed and self-built indicators in every regression, not shown. (2)–(4) from 2009–11; (6)–(7) to 2011–13 (import-share flag omitted: equals self-built). SE clustered by firm; province bootstrap p in brackets. No controls: survey-pair FE only. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Summary of the Evidence

FAT: 19,332 firms, 15 countries. **Vietnam SME panel**: 6,913 firm-rounds, 2011–2015; *all used* $=\geq 95\%$ of equipment bought used (10.4%). Vietnam variables demeaned within sector $\times$ province $\times$ round.

	What I ran	Result
1	**FAT**: LPM of $\mathbf{1}\{\text{used only}\}$ on capability, size, trade, finance; sector, region FE	College manager -6.8 pp, KPIs -1.7 pp, credit-constrained $+3.0$ pp (mean 12.5%)
2	**FAT**: technology / TFP on used-only and both dummies, same controls	Tech. score -0.52 (mean 6.46); revenue TFP -0.14
3	**VN**: capability and size gap, all used vs. others, within cells	Capability -0.45 s.d., size -0.39 s.d.; gap in ICT, manager, accounts, not workers' schooling
4	**VN**: machine age on capability, by group	$+1$ s.d. capability $\Rightarrow$ machines 1.1 yrs younger (all used), 0.4 (any new)
5	**VN**: LPM of all used on log assets, log employment, capability	Assets -0.006 (n.s.) **once size fixed**; capability -0.031^{***}
6	**VN**: value added per machine dollar; TFPR gap by capability tercile	VA per machine \$ gap $+0.002$ (0.050); TFPR gap ≈ -0.14 in every tercile

Capability, not wealth, sorts firms into used capital;
all-used firms are not short of capital.

Model

Model in One Slide: New Capital Carries the Technology

Firm $(z,\ x,\ a)$: productivity, capability, wealth. Lucas span-of-control; rents capital K at R per dollar.

Choice: used or new capital. A dollar of either gives the same capital services; **new capital embodies the new technology** ϕ:

$$\text{used: } Y = e^{z}\left(L^{\alpha}K^{1-\alpha}\right)^{\eta}$$
$$\text{new: } Y = e^{z+\phi}\left(L^{\alpha}K^{1-\alpha}\right)^{\eta}$$

Know-how: to use that technology, pay up front

$$F = \kappa\, e^{-\tilde{x}}\, e^{z/(1-\eta)}, \qquad \tilde{x} = x - \sigma e_F$$

Money: collateral constraint $K \le \theta a$ (used), $K + F \le \theta a$ (new)

Two reasons to stay with used capital

- **Capability-bound**: $\tilde{x} < \bar{x}$, would stay used even with unlimited credit
- **Finance-bound**: $\tilde{x} \ge \bar{x}$ but wealth too low for its size

Calibrated to 10 Vietnam moments

All-used firms: capability-bound	**96.4%**
finance-bound (bootstrap 90%: 0–19%)	3.6%
Adoption share, baseline	93.1%
no collateral constraint	93.3%
capability → median adopter	**97.3%**

Mostly a know-how barrier: capability lifts adoption 17× more than credit.

Calibration Targets: Ten Vietnam Moments

Parameter	Meaning	Value	Target (data value)
From the JMP: $\alpha = 2/3$, $\eta = 0.79$, $W = 0.540$, $R = 0.119$			
σ_ℓ	employment noise	0.425	var. minus autocov. of log employment (0.181)
σ_z	productivity dispersion	0.176	2-round autocov. of log employment (0.734)
ρ	capability–productivity	0.756	corr(index, log employment) (0.61)
ϕ	gain from new technology	0.022	size gap, all used − others (-0.39 s.d.)
$\ln(\kappa/W)$	adoption cost level	-5.02	all-used share (0.104)
σ	know-how not in the index	1.59	capability gap (-0.45 s.d.)
θ	collateral	11.7	all used on log assets, size fixed (-0.006)
μ_a	wealth level	1.78	debt / value added (0.60)
β_a	wealth–size	1.13	corr(log assets, log employment) (0.64)
σ_a	wealth dispersion	1.58	variance of log assets (1.73)
Not targeted: VA per machine $ gap: model $+0.010$, data $+0.002$ (0.050); credit-constrained → used only: model $+1.4$ pp, FAT $+3.0$ pp			
Not matched: VA per worker gap 0.00 vs. -0.56; machinery per worker -0.01 vs. -0.59; TFPR -0.08 vs. -0.21			

1,000,000 simulated firms; nine moments matched within 0.0009 s.e. (smoothed choice), 0.007 (exact); 60-draw bootstrap.

The model fits the capital side it was not targeted to, and misses the labor side.